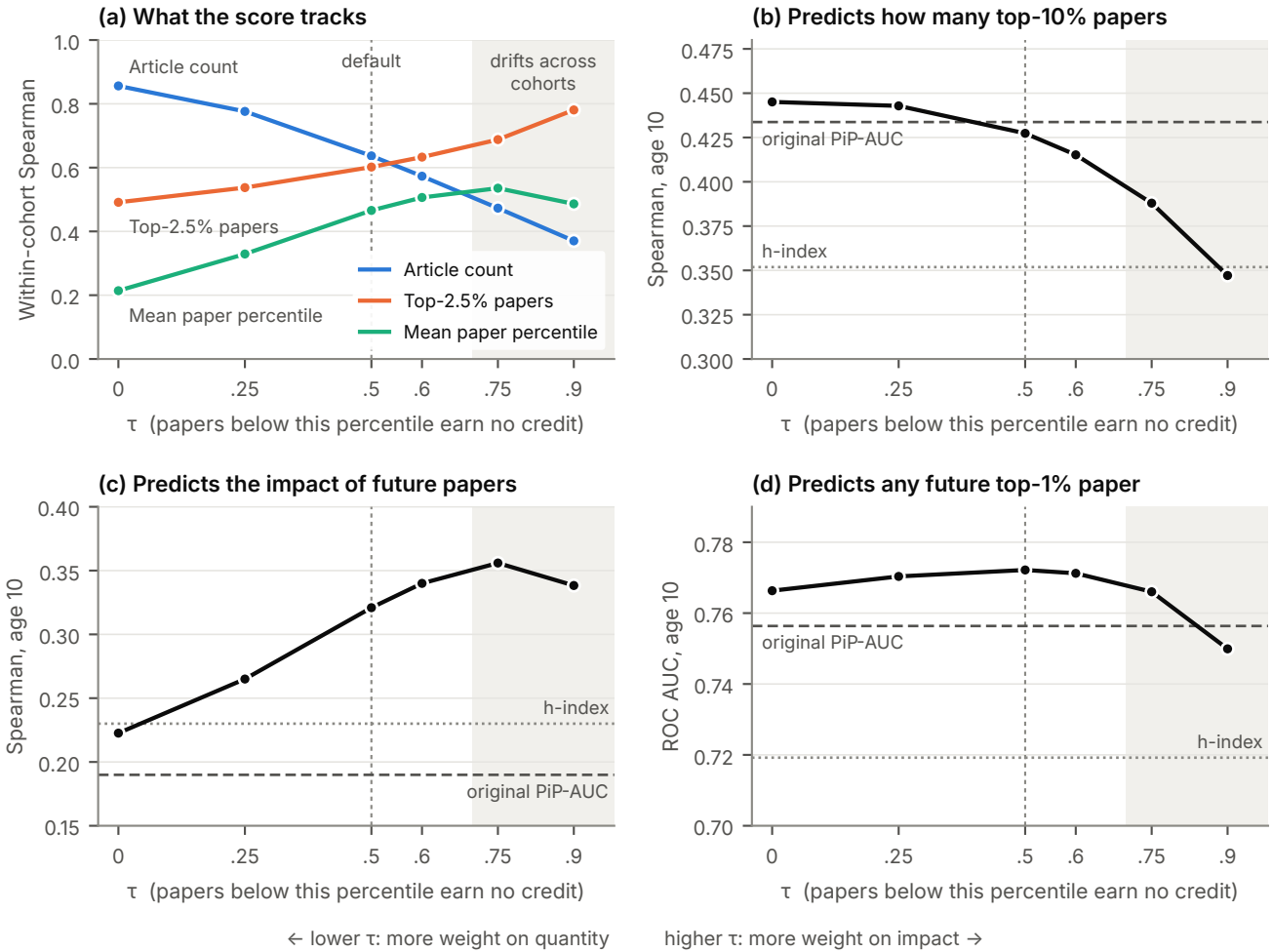
(a) What the score tracks
Within-cohort Spearman
1.0
0.8
0.6
0.4
0.2
0.0
Article count
default
drifts across cohorts
Top-2.5% papers
Mean paper percentile
Article count
Top-2.5% papers
Mean paper percentile
0
.25
.5
.6
.75
.9
τ (papers below this percentile earn no credit)
(b) Predicts how many top-10% papers
Spearman, age 10
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0.300
original PiP-AUC
h-index
0
.25
.5
.6
.75
.9
τ (papers below this percentile earn no credit)
(c) Predicts the impact of future papers
Spearman, age 10
0.40
0.35
0.30
0.25
0.20
0.15
h-index
original PiP-AUC
0
.25
.5
.6
.75
.9
τ (papers below this percentile earn no credit)
(d) Predicts any future top-1% paper
ROC AUC, age 10
0.78
0.76
0.74
0.72
0.70
original PiP-AUC
h-index
0
.25
.5
.6
.75
.9
τ (papers below this percentile earn no credit)
← lower τ: more weight on quantity
higher τ: more weight on impact →